

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 12

ODD OR EVEN

AIM: To write and execute an 8085 assembly language program to find odd or even and store the result in memory.

ALGORITHM

1. Load the number into accumulator A.
2. Rotate A right using RRC.
3. Check the Carry flag.
4. If Carry = 0, the number is even.
5. If Carry = 1, the number is odd.
6. Stop the program.

CODE:

```
MVI A,25H
```

```
RRC
```

```
JC ODD
```

```
MVI A,00H
```

```
STA 8050H
```

```
HLT
```

```
ODD: MVI A,01H
```

```
STA 8050H
```

```
HLT
```

OUTPUT:

FileResetAssemblerDebugHelp

Registers

A01

BC0000

DE0000

HL0000

PSW0000

PC42DE

SPFFFF

Int-Reg00

Flag

S0

Z0

AC1

P0

C0

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2500

-

+

05

Update Memory

Load me at

1LDA 2500

2ANI 01

3JZ EVEN

4

5MVI A, 01

6STA 2501

7HLT

8

9EVEN: MVI A, 00

10STA 2501

11HLT

DataStackKeyPadMemoryI/O Ports

Start2500

OK

Address (Hex)AddressData

09C425005

09C525011

09C625020

09C725030

09C825040

09C925050

09CA25060

09CB25070

09CC25080

09CD25090

09CE25100

09CF25110

09D025120

09D125130

Line NoAssembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG

IN

11:50

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